Rory Byrne

rory.byrne.24@ucl.ac.uk | linkedin.com/in/rory-byrne1 | github.com/rorybyrne1 | rorybyrne.com

Education

University College London — London, United Kingdom
MEng Computer Science — *Sep. 2024 – June 2028*

- Current weighted average of 83.21% across my first two years

Wheatley Park School — Oxford, United Kingdom
A-Levels and GCSEs — *Sep. 2017 – June 2024*

- Achieved A*A*A* in Mathematics, Computer Science, Physics, and A in Further Mathematics AS

Experience

Software Engineering Intern — July – Sep. 2026
MotionInput Games — *London, United Kingdom*

- Developed a real-time modular puppeteering system in Python using pose landmarks to drive custom 2D avatars
- Implemented landmark smoothing, jump detection and missing-landmark handling for tracking stability
- Implemented OpenCV and PySide6 renderers for in-window and desktop overlay display with configurable assets
- Debugged, packaged, and shipped four MotionInput games to the Microsoft Store using MSIX

Web Developer — June 2025 – Present
Solas Oxford CIC — *Oxford, United Kingdom*

- Leading the redesign and rebuild of a live WordPress website to improve organisation and maintainability
- Developing a reusable system for trainings and events to centralise content and reduce redundancy
- Translating visual designs into responsive, accessible page layouts in collaboration with a graphic designer

Waiter — June - Sep. 2024, 2025
The Seven Stars at Marsh Baldon, The Mole Inn — *Oxford, United Kingdom*

- Worked effectively in a fast-paced environment, managing multiple tables and orders during busy service periods
- Collaborated closely with other waiters and kitchen staff to ensure smooth operations and timely service

ISIS Synchrotron Group Work Experience — June – July 2023
Rutherford Appleton Laboratory — *Didcot, United Kingdom*

- Developed Python GUIs to help accelerator physicists monitor and understand proton beams
- Implemented NumPy-based peak and loss finders for live MQTT beam-intensity monitor data

Projects

ReadingStar 2.0 | *Python, FastAPI, JavaScript* — Oct. 2025 – March 2026

- Built an offline AI-powered educational app using music to support speech development for autistic children
- Developed a FastAPI backend to manage songs, lyrics, and user progress with local data persistence
- Designed and implemented an accessible, low-friction UI flow for creating and singing along to songs
- Collaborated with **Intel**, **IBM**, the **National Autistic Society**, and **MotionInput Games**, delivering regular progress updates and discussing design decisions with members of each organisation

Twin-Stick Shooter Game | *Python, SQL* — Sep. 2023 - March 2024

- Developed a twin-stick shooter game in Python using Pygame where the player must defeat hordes of enemies
- Implemented user accounts, multiplayer, character customisation, and a leader board using a MySQL database

Awards and Achievements

- Achieved 100th percentile in the UCL STAT admissions test out of over 2000 shortlisted applicants
- Best in School in Senior Maths and Elite Bebras Computing Challenges 2022 and 2023
- Achieved 10th in the UK in the Náboj International Mathematics Competition 2023 working in a team of 5
- Achieved runners up out of over 1200 schools in the Ritangle 2023 competition working with my peers

Technical Skills and Interests

Programming Languages: Python, C, C++, C#, Java, JavaScript, HTML, SQL, Haskell, GLSL
Graphics & Game Development: OpenGL, Unity, Godot
Libraries & Frameworks: NumPy, MySQL, Matplotlib, FastAPI, Electron
Developer Tools: Git, GitHub, Jupyter, VS Code, Postman, WordPress, LaTeX
Interests: Music, Photography, Woodworking and Wood Carving, Badminton